

Ashutosh Trivedi

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Research Profile

I develop mathematical foundations and tools for **trustworthy AI systems that learn and make decisions under uncertainty while satisfying formal guarantees**. My research brings together reinforcement learning, formal verification, and symbolic reasoning, with applications to safety-critical control and accountable software.

Research themes: reinforcement learning and sequential decision-making; logic, automata, and formal verification; stochastic systems and control; fairness and accountability; learning-enabled cyber-physical and medical systems.

Appointments & Education

CU Boulder · Associate Professor *2023–present*
Assistant Professor, 2017–2023; Research faculty, 2015–2017.
IIT Bombay · Assistant Professor *2012–2015*
University of Pennsylvania · Postdoc *2011–2012*
University of Oxford · Postdoc *2009–2010*
University of Warwick · Ph.D., CS *2009*

Selected Recognition

DARPA NeuS Disruptive Idea Award 2025
Royal Society Wolfson Visiting Fellow 2024–26
CAV Distinguished Paper Award 2024
NSF CAREER Award 2022
ACM Senior Member 2024
IEEE Senior Member 2025
HSCC Best Paper Award 2012

Selected Research Support

- **NSF CAREER** (2022–27): reinforcement learning for recursive MDPs and beyond.
- **NSF DASS** (2023–26): accountability of tax-preparation software.
- **DARPA / NeuS** (2025–26): control transfer via neural simulation functions.
- **AB Nexus** (2024–26): AI-optimized cardiac pacing with CU Anschutz.
- **NSF CCF** (2020–23): omega-regular objectives for model-free reinforcement learning.
- **NSF EPCN** (2020–23): secure-by-construction controller synthesis for cyber-physical systems.

Selected Professional Service

- **Action Editor**, Transactions on Machine Learning Research.
- **Editorial Board**, Information Processing Letters.
- **General Co-Chair**, VMCAI 2025.
- **Co-Chair**, SNR 2022 and 2026.
- Senior reviewing and program-committee service across CAV, ICALP, AAAI, NeurIPS, and ICLR.

Selected Publications

Regular Reinforcement Learning. T. Dohmen, M. Perez, F. Somenzi, A. Trivedi. *CAV*, 2024. *Distinguished Paper Award*.

Omega-Regular Decision Processes. E. M. Hahn, M. Perez, S. Schewe, F. Somenzi, A. Trivedi, D. Wojtczak. *AAAI*, 2024.

Recursive Reinforcement Learning. E. M. Hahn, M. Perez, S. Schewe, F. Somenzi, A. Trivedi, D. Wojtczak. *NeurIPS*, 2022.

Correct-by-Construction Reinforcement Learning of Cardiac Pacemakers from Duration Calculus Requirements. K. Dole, A. Gupta, J. Komp, S. N. Krishna, A. Trivedi. *AAAI*, 2023.

Stochastic Neural Simulation Relations for Transferring Control under Uncertainty. A. Nadali, A. Trivedi, M. Zamani. *NeuS*, 2025. *Disruptive Idea Award*.

On the Robustness of Fairness Practices: A Causal Framework for Systematic Evaluation. V. Monjezi, A. Kumar, A. Trivedi, G. Tan, S. Tizpaz-Niari. *ICSE*, 2026.

Information-Theoretic Testing and Debugging of Fairness. V. Monjezi, A. Trivedi, G. Tan, S. Tizpaz-Niari. *ICSE*, 2023.

Regular Transformations of Infinite Strings. R. Alur, E. Filiot, A. Trivedi. *LICS*, 2012.

Optimal Scheduling for Constant-Rate Multi-Mode Systems. R. Alur, A. Trivedi, D. Wojtczak. *HSCC*, 2012. *Best Paper Award*.

Assume-Guarantee Reinforcement Learning. M. Kazemi, M. Perez, F. Somenzi, S. Soudjani, A. Trivedi, A. Velasquez. *AAAI*, 2024.

A PAC Learning Algorithm for LTL and Omega-Regular Objectives in MDPs. M. Perez, F. Somenzi, A. Trivedi. *AAAI*, 2024.

Neural Closure Certificates. A. Nadali, V. Murali, A. Trivedi, M. Zamani. *AAAI*, 2024.

An LLM Agentic Approach for Legal-Critical Software: A Case Study for Tax Preparation Software. S. Gogani-Khiabani, A. Trivedi, D. Saha, S. Tizpaz-Niari. *ICSE*, 2026.

Metamorphic Testing and Debugging of Tax Preparation Software. S. Tizpaz-Niari, M. Wagner, S. Darian, K. Reed, A. Trivedi. *ICSE-SEIS*, 2023.